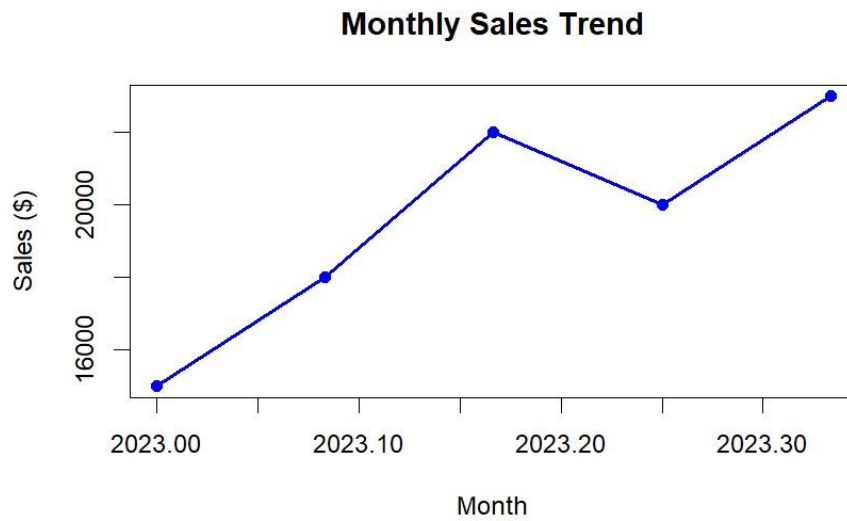


SET-18

QUESTION-1



CODE:

```
# Monthly Sales Data sales <- c(15000, 18000, 22000,
20000, 23000)

# Create time series sales_ts <- ts(sales,
start = c(2023, 1),

frequency = 12)

# Plot time series

plot(sales_ts, type = "o", col = "blue",

pch = 16,

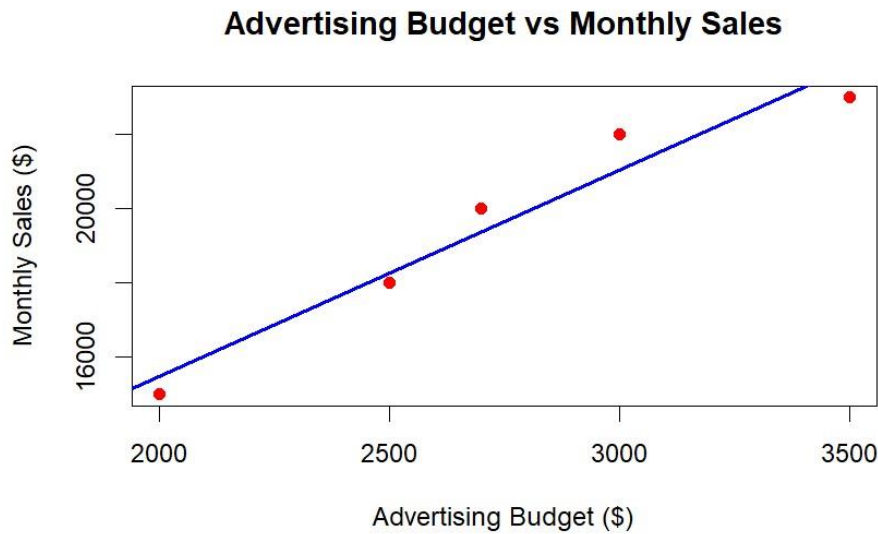
lwd = 2,

xlab = "Month",

ylab = "Sales ($)",

main = "Monthly Sales Trend")
```

QUESTION-2



CODE:

```
# Advertising Budget advertising <- c(2000, 2500, 3000, 2700,
3500)

# Monthly Sales sales <- c(15000, 18000, 22000, 20000, 23000)

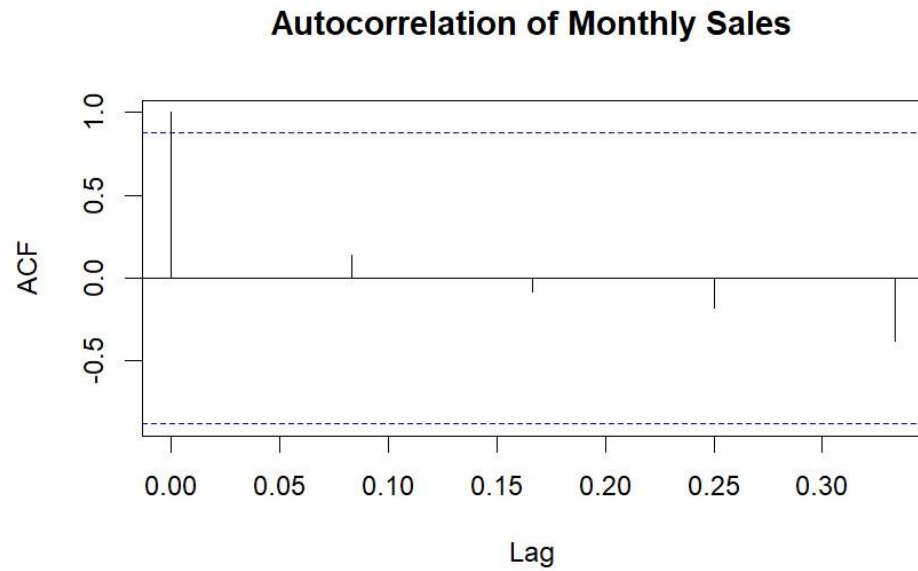
# Scatter Plot

plot(advertising,
sales,    pch = 19,    col = "red",    xlab = "Advertising Budget
($)",    ylab = "Monthly Sales ($)",    main = "Advertising Budget
vs Monthly Sales")

# Regression Line

abline(lm(sales ~ advertising),    col = "blue",
lwd = 2)
```

QUESTION-3



CODE:

```
# Monthly Sales

Data sales <- c(15000, 18000, 22000, 20000, 23000)

# Create Time Series

sales_ts <- ts(sales, start = c(2023, 1),

frequency = 12)

# Autocorrelation Plot

acf(sales_ts, main = "Autocorrelation of Monthly Sales")
```